AWS Certified Cloud Practitioner

Welcome & Exam Overview

- **Entry-level certification:** Validates foundational knowledge of AWS cloud concepts for both technical and non-technical roles.
- **Global recognition:** Widely accepted as a credential to enhance career prospects in cloud computing and IT.
- **No prerequisites:** Open to anyone, regardless of technical background, making it accessible to all professionals.



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AWS Cloud Practitioner Exam Details

Format, Cost & Domains

- **Exam format:** 90-minute exam with multiple-choice and multiple-response questions.
- **Passing score & cost:** Score 700/1000 to pass; exam costs approx. $100 USD.
- **Domains covered:** Cloud concepts, Security, Technology, Billing & Pricing.



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Cloud Computing Basics

On-demand, Scalable & Cost-Efficient

- **On-demand delivery:** Access IT resources like compute, storage, and networking anytime via the internet.
- **Scalability & elasticity:** Easily scale resources up or down based on workload demands.
- **Pay-as-you-go model:** Only pay for what you use, eliminating costly upfront hardware investments.



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Benefits of Cloud Adoption

Why Organizations Choose AWS



Cost savings
Eliminates capital expenses of physical data centers; pay only for resources used.



Elasticity & agility
Scale resources up or down instantly to match business demand.



Global reach
Deploy applications and services worldwide with low latency.



Enhanced security
AWS provides encryption, compliance, and enterprise-grade protections.

AWS Global Infrastructure

Regions, Availability Zones & Edge Locations



Regions
Geographically isolated areas hosting multiple Availability Zones.



Availability Zones (AZs)
Independent data centers with power, cooling, and networking for fault tolerance.



Edge locations
Global network points enabling low-latency content delivery through CloudFront.

AWS Compute Services

Scalable Compute Options

- **Amazon EC2:** Virtual servers with flexible configurations, pricing models, and integrations.
- **AWS Lambda:** Serverless compute for event-driven workloads; pay only for execution time.
- **Elastic Beanstalk:** PaaS for deploying and managing applications without infrastructure management.
- **Containers:** ECS, EKS, and Fargate for orchestrating and scaling containerized applications.

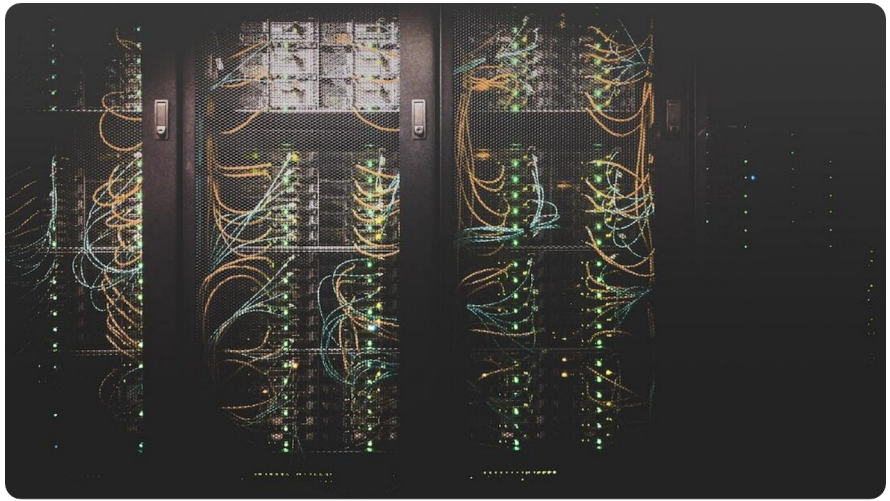


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AWS Storage Services

Flexible Data Storage Options



Amazon S3
Highly durable object storage for files, media, and backups.



Amazon EBS
Block storage volumes for persistent EC2 data with snapshot support.



Amazon EFS
Scalable file storage shared across multiple EC2 instances.



Amazon Glacier
Low-cost archival storage for long-term data retention.

AWS Database Services

Relational & NoSQL Options



Amazon RDS
Managed relational databases (MySQL, PostgreSQL, Oracle, SQL Server, MariaDB).



Amazon DynamoDB
Fully managed NoSQL database with single-digit millisecond performance.



Read Replicas
Scale read-heavy workloads and enable disaster recovery across regions.



Security & scaling
Built-in encryption, multi-AZ deployments, and automated backups.

AWS Networking & Content Delivery

Connecting and Distributing Applications



Amazon VPC
Create isolated virtual networks with full control over IP ranges and routing.



Elastic Load Balancing (ELB)
Distributes traffic across multiple resources for high availability.



Amazon Route 53
Highly available DNS service with domain registration and traffic routing policies.



Amazon CloudFront
Content delivery network (CDN) reducing latency with global edge locations.

AWS Messaging & Integration

Decoupling and Event-Driven Architectures



Amazon SQS
Fully managed message queuing to decouple distributed systems.



Amazon SNS
Pub/sub service to broadcast messages to multiple endpoints (email, SMS, Lambda).



Integration
Both services integrate with Lambda and other AWS tools for automation.



Scalability
Can handle millions of messages per second, ensuring resilience.

AWS Security & Identity

Protecting Access and Data

- **IAM (Identity & Access Management):** Manage users, groups, and roles with fine-grained permissions.
- **Shared Responsibility Model:** AWS secures the cloud; customers secure their applications and data.
- **MFA & Access Keys:** Multi-factor authentication adds an extra layer of security for accounts.
- **Advanced protections:** AWS WAF and Shield defend against DDoS and web exploits.

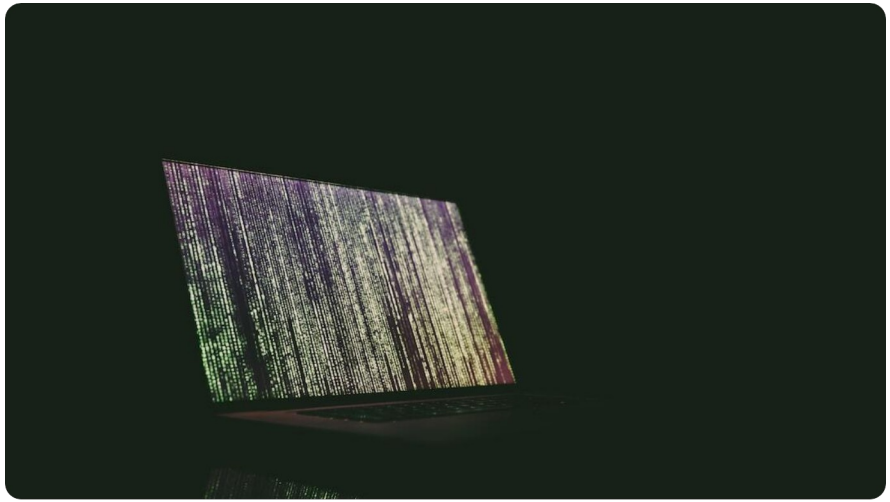


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AWS Monitoring & Logging

Visibility and Operational Insights

- **Amazon CloudWatch:** Collects and monitors metrics, logs, and alarms for AWS resources and apps.
- **AWS CloudTrail:** Provides audit logs of API activity across AWS accounts.
- **Trusted Advisor:** Offers real-time guidance on cost optimization, security, and performance.



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AWS Automation & Deployment

Accelerating DevOps Workflows

- **AWS CloudFormation:** Infrastructure as Code for consistent and automated provisioning.
- **CodePipeline:** Continuous integration and continuous delivery (CI/CD) automation.
- **CodeCommit, CodeBuild, CodeDeploy:** Source control, build automation, and deployment services for DevOps.
- **Agility & consistency:** Automates repeatable tasks and reduces human error in deployments.



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AWS Governance & Multi-Account Management

Centralized Control & Compliance



AWS Organizations
Manage multiple AWS accounts centrally with consolidated billing and policies.



Service Control Policies (SCPs)
Set permission boundaries across accounts to enforce compliance.



AWS Control Tower
Automates setup of secure multi-account environments with best practices.



Centralized governance
Improves visibility, cost management, and security across large organizations.

AWS Pricing Models

Flexible Cost Structures



On-Demand Instances
Pay by the hour/second with no upfront costs or long-term commitments.



Reserved Instances
Up to 72% savings with 1- or 3-year commitments for predictable workloads.



Spot Instances
Bid for unused capacity at up to 90% discount, ideal for flexible workloads.



Savings Plans & Free Tier
Flexible pricing for consistent use and free tier for new customers.

AWS Cost Optimization Tools

Controlling and Reducing Expenses

- **AWS Budgets:** Set custom cost and usage budgets with alerts for overspending.
- **Cost Explorer:** Visualize, analyze, and forecast spending trends across services.
- **Trusted Advisor:** Identifies unused resources and recommends optimizations for cost, security, and performance.
- **Resource tagging:** Track costs by team, project, or department with detailed tagging strategies.



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AWS Well-Architected Framework

Five Pillars of Best Practices

- **Operational Excellence:** Focus on operations automation, monitoring, and continuous improvement.
- **Security:** Protect information and systems with encryption, identity, and compliance.
- **Reliability:** Design fault-tolerant systems with automated recovery and scaling.
- **Performance Efficiency:** Use scalable, cost-efficient, and evolving technologies.
- **Cost Optimization:** Eliminate unnecessary expenses and optimize resource usage.

AWS Disaster Recovery Strategies

Ensuring Business Continuity

- **Backup & Restore:** Simplest and cost-effective strategy; data is backed up and restored when needed.
- **Pilot Light:** Minimal version of environment always running; scaled up during disaster.
- **Warm Standby:** Smaller-scale environment running in parallel; can be scaled up quickly.
- **Multi-Site (Active-Active):** Fully redundant environments in multiple regions; highest availability.



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AWS Compliance & Governance

Standards and Certifications

- **Global standards:** AWS complies with ISO 27001, SOC 1/2/3, PCI DSS, HIPAA, and GDPR.
- **AWS Artifact:** Provides on-demand access to compliance reports and security documentation.
- **Shared compliance:** AWS manages infrastructure compliance; customers manage application-level compliance.
- **Industry trust:** Enables regulated industries like healthcare and finance to adopt AWS confidently.

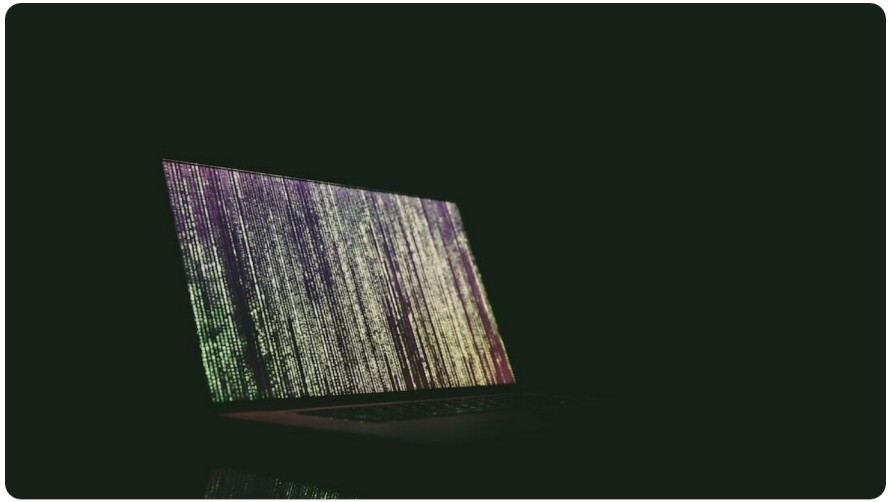


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AWS Exam Preparation Resources

Study Smarter, Not Harder



AWS Whitepapers & FAQs
Official guides covering cloud concepts, security, and billing.



Sample questions & practice exams
Familiarize yourself with exam format and difficulty.



AWS Free Tier practice
Hands-on labs to reinforce learning without extra cost.



Community & study groups
Engage with peers, forums, and AWS Learning Communities.

AWS Support Plans

From Basic to Enterprise

- **Basic:** Free support with 24/7 access to documentation, whitepapers, and forums.
- **Developer:** Email support for testing and development, response time under 24 hours.
- **Business:** 24/7 support with phone, chat, and guidance for production workloads.
- **Enterprise:** Dedicated Technical Account Manager (TAM) and concierge support.

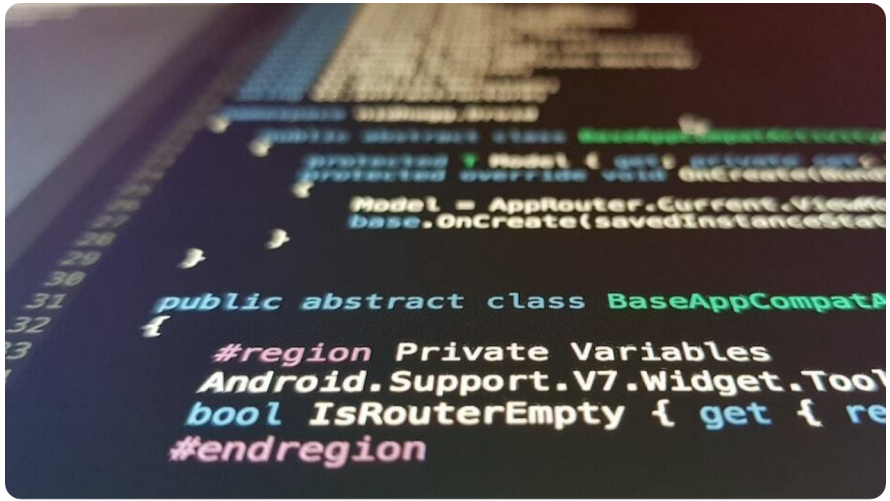


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AWS Marketplace

Third-Party Software & Integrations

- **Wide range of products:** Offers thousands of pre-configured third-party software solutions.
- **Seamless integration:** Easily deploy software directly into AWS environments.
- **Flexible pricing:** Pay-as-you-go or annual contracts, integrated with AWS billing.
- **Security & compliance:** All solutions vetted for security and enterprise-readiness.



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Final Revision & Success Tips

Maximize Your Exam Readiness

- **Focus on key domains:** Revisit Cloud Concepts, Security, Technology, and Billing.
- **Practice regularly:** Use sample questions, flashcards, and mock exams to reinforce knowledge.
- **Hands-on labs:** Leverage AWS Free Tier to gain practical experience with core services.
- **Exam strategy:** Manage time, read questions carefully, and eliminate wrong options.



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AWS Cloud Practitioner Cheat Sheet

High-Yield Exam Revision

- **Core Domains:** Cloud Concepts, Security, Technology, Billing & Pricing.
- **Key Services:** EC2, Lambda, S3, RDS, DynamoDB, VPC, IAM, CloudFront, Route 53.
- **Pricing Models:** On-Demand, Reserved, Spot, Savings Plans, Free Tier.
- **Best Practices:** Well-Architected 5 Pillars + Disaster Recovery (Backup, Pilot Light, Warm Standby, Multi-Site).
- **Exam Tips:** Practice mock tests, focus on keywords, manage time, eliminate wrong answers.